Project Documentation: Distribution Panel – Phase 1 Completion

Project: GE_Wiring
Sections: Distribution Panel | Operating Panel | Soft Starter
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Logical Block Diagram / Concept Map

Power Sources → Power selection and Protection → Distribution & conversion → Loads & Interfaces

Block-1
- Electric Generator (3ϕ, 440v, 50Hz)
- shore power (3ϕ, 440v, 50Hz)

Blk-2
- CO-SW1

Blk-3
- cooling System
- Compressor
- GEN5 cart
- Light System
- AC-DC conv.

Blk-4
- throught loads to Operatin panel & Emerge Sw

J_GEN
J_Shore
W_SRC_GEN
W_SRC_SHORE
Block-1

X_CO_SW1
W_MAIN_DIST
Block-2

CB_COMPRESSOR
CB_GEN5
CB_LIGHT
MOD_AC_DC
W_FED_*
W_LIGHT_AC
W_BEACON_DC
Block-3

X_OP_PANEL
M_COOLING
M_COMPRESSOR
M_GEN5_CART
L_BEACON
SW_EMERGENCY
Block-4

January 2026 | Thursday 22

Block 1 ⇒ electric generator & shore power

Electric Generator
3ϕ 440V 50HZ

L1 → 1
L2 → 5
L3 → 9
N → 13
} CO-SW1
PE → Earth GND

Shore Power
3ϕ 440V 50HZ

L1 → 3
L2 → 7
L3 → 11
N → 15
PE → Earth GND

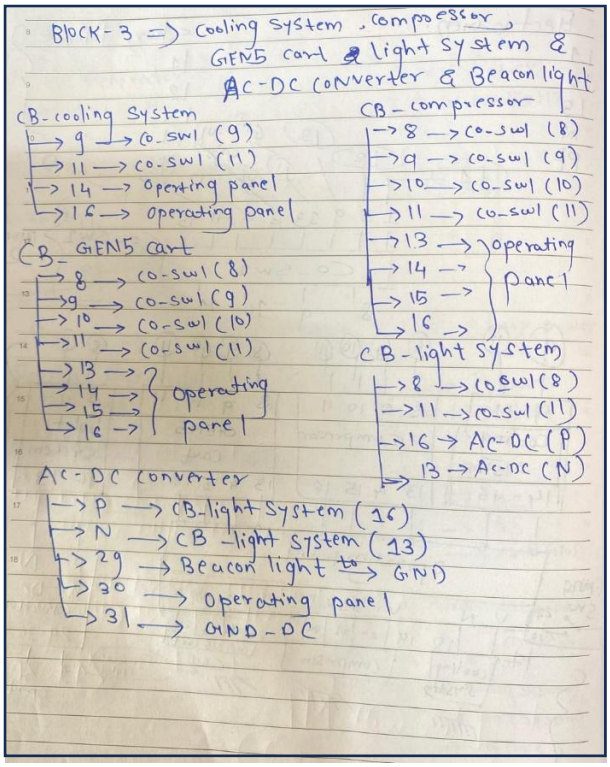
Block 2 ⇒ CO-SW1 & voltmeter & ~~Power Sources~~

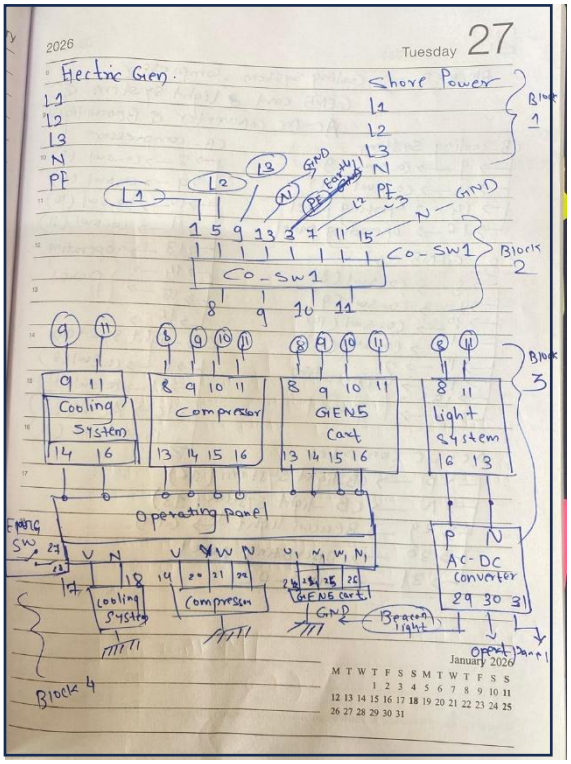
1 → L1
5 → L2
9 → L3
13 → N
} electric Gen

3 → L1
7 → L2
11 → L3
15 → N
} Shore power

→ N → 8
→ POWER-L → 9
} CO-SW1
→ V1 → 9 (CB_cooling system)
→ V2 → 11 (CB-cooling system)
→ V3 → 11 (GEN 5 cart)
→ N → 8 (CO-SW1)

What looks like the end of the road is more often just a bend in the road.





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1. Objective & Approach

The overall project consists of **three major electrical sections**:

1. Distribution Panel
2. Operating Panel
3. Soft Starter

To reduce design risk and ensure correctness, I adopted a phased development approach, completing and validating one section at a time instead of modelling all sections in parallel.

The Distribution Panel was selected as Phase 1, as it forms the electrical foundation for the remaining sections.

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2. Phase 1 Scope: Distribution Panel

The Distribution Panel was analysed and implemented as a standalone logical system, independent of downstream control logic.

System Decomposition

Before any software modelling, the panel was decomposed into four logical blocks:

1. **Power Sources**

    - Electric Generator (3-phase)
    - Shore Power (3-phase)
2. **Power Selection**
    - Changeover Switch (GEN / SHORE)
3. **Distribution & Conversion**
    - Main distribution wiring
    - Individual circuit breakers (Cooling, Compressor, GEN5 Cart, Lighting)
    - AC–DC conversion for beacon/light system
4. **Interfaces & Loads**
    - Operating Panel interface terminals
    - Cooling system
    - Compressor
    - GEN5 Cart
    - Beacon and emergency loop

This block-level thinking ensured the wiring logic was correct before translating it into a digital schematic.

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3. Implementation Method

The validated logical model was then implemented using Wire Viz (v0.4.1) to generate a digital Electrical Interconnect Diagram (EID).

Key implementation principles:

- Correct modelling of 3-phase + neutral loads
- Explicit separation between distribution logic and external loads
- Operating Panel treated as an interface boundary, not a load
- Safety circuits (emergency loop) isolated and clearly defined

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5. **Validation & Refinement**
   During implementation, several iterations were required to align the digital model with physical reality:

- Corrected initial simplifications (single-phase assumptions for 3-phase motors)

- Introduced Operating Panel as a central routing interface
- Ensured all source inputs were explicitly connected
- Refined pin allocation to improve safety and clarity

The final output accurately represents:

- Source → Selection → Protection → Distribution → Interface flow
- Practical wiring paths suitable for manufacturing and maintenance

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5. Phase 1 Status

Distribution Panel: Completed

- Logical architecture finalized
- Digital schematic generated
- Ready for review and discussion

This section is now stable and can be used as a reference baseline for the remaining system sections.

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6. Proposed Next Steps

I propose to follow the same phased methodology for the remaining sections:

Phase 2: Operating Panel

Phase 3: Soft Starter

Each section would be:

1. Logically decomposed
2. Independently modeled
3. Reviewed before proceeding further

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7. Request for Confirmation

Please confirm if this section-by-section approach is acceptable, and if I may proceed with:

- Operating Panel as the next phase